U.S. CITIZEN

Griffin DesRoches

Cambridge, MA | gmdesroches@umass.edu | griffindesroches.me | griffindesroches | griffinxaxa

Summary

Software engineer with a Statistics and Mathematics double major and Computer Science minor, experienced in building and improving defense-quality production microservices, CI/CD pipelines, automated testing, and full-stack applications. Proficient in C++, Python, Java, JavaScript, and Linux; amongst other skills in R and data modelling/analysis. seeking backend, systems, infrastructure, or mission-critical software engineering roles for Summer 2027.

Education

University of Massachusetts Amherst *Expected May 2028*

B.S. in Statistics and Data Science; B.S. in Mathematics; Minor in Computer Science

- **Relevant coursework:** Data Structures, Object-Oriented Programming, Scientific Computing, Advanced Linear Algebra, Data Science, Introductory Machine Learning, Probability, Statistical Inference, Real Analysis, Calculus

Cambridge Rindge & Latin School *Sept 2021 – May 2025*

Experience

Software Engineer Intern *Maynard, MA*

Fisheye Software, Inc. *May 2026 – Aug 2026*

- Expanded automated unit-test coverage across a production microservice platform from 30% to 75%, identifying high-risk components, implementing tests, and creating CLI tooling to prioritize files below coverage thresholds.
- Replaced fixed-delay synchronization with event-driven completion signaling in the CI/CD test pipeline, reducing runtime by 37% and eliminating 30–35 minutes of idle time per run, Saving 40 hours of pipeline runtime weekly.
- Diagnosed and shipped a production bug fix within one day by correcting microservice status-handling logic that incorrectly reported successful jobs as failures, enabling a successful stakeholder demo.
- Designed a parallel unit-test execution strategy across two EL9 GitLab runners, documenting required infrastructure changes and authoring implementation stories to halve CI pipeline runtime.

Research Intern – Statistical Software *Boston, MA*

Dana-Farber Cancer Institute, Parmigiani Lab *Feb 2025 – May 2025*

- Developed an interactive R/Shiny interface for the open-source `DRCMD` package, integrating configurable modeling workflows, cross-fitting, trimming, and real-time diagnostic outputs.
- Built reproducible simulation and validation pipelines to test package behavior across missing-data scenarios and multiple machine-learning backends, contributing to a submitted software publication.
- Conducted statistical analysis of missing data patterns across five clinical variables, evaluating Random Forest and Gradient Boosting models for treatment-effect estimation under high missingness and noise, validating the doubly robust properties of the DRCMD R package through simulation.

Software Engineer Intern *Cambridge, MA*

Innovators for Purpose *Jul 2023 – Nov 2023*

- Designed and built a Node.js/Express MVC backend with a relational database schema of 30+ models and 50+ relationships, powering a full-stack classroom research application used by a school district.
- Collaborated within a five-person Agile team, using Git/GitHub for version control to integrate frontend and backend features across sprints while maintaining a stable, regression-free codebase.

Projects

CS2 Match Outcome Prediction *Github repository*

- Scraped, cleaned, and engineered a dataset of 8,000+ professional esports matches in Python, constructing 5-game rolling performance averages to capture temporal team form.
- Trained and evaluated logistic regression and tree-based ensemble models with the R RF library, achieving 65–67% accuracy on time-aware validation while analyzing feature importance and model drift.

easytree.xyz - ASCII File Directory Generator *Github repository*

- Built and deployed a full-stack web application (React, Node.js/Express) that generates ASCII file trees from public GitHub repositories, streamlining documentation, onboarding, and LLM-assisted code analysis.

Personal site - Griffindesroches.me *Github repository*

- Designed and developed a responsive personal website using HTML5 and CSS3, serving as a portfolio to showcase project work as well as a blog. Deployed the site on Vercel for simple deployment, updates, and hosting.

Technical Skills

Languages: C++, Python, Java, JavaScript, SQL, R

Frameworks and libraries: Node.js, Express, React, R/Shiny, pandas, NumPy

Systems and tools: Linux, Git, GitHub, GitLab, CI/CD, REST APIs, ZeroMQ, Bash, VS Code, IntelliJ, LaTeX

Miscallaneous: Unit Testing, Code Coverage, Agile/Scrum, Stakeholder Reporting, Pair Programming,